

Special Relativity

Lecture 6: Tensors and the Relativistic Lorentz Force

Lectures by Leonard Susskind

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Chapter 1

Tensors and the Relativistic Lorentz Force

Tonight we turn from scalar fields to electrodynamics. The preceding discussion established a useful pattern: one action describes both how a field acts on a particle and how the particle acts back on the field. We shall eventually apply that pattern to the electromagnetic field, but first we need notation strong enough to expose the symmetry of the calculation.

This preliminary work is not decoration. A minus sign, a zero, an equals sign, and vector notation each compress an idea that would otherwise have to be repeated in words. Tensor notation does the same for relativity. Once upper and lower indices, contractions, and transformation laws are in place, the electric and magnetic forces will assemble themselves into one four-dimensional equation.

1.1 Four-Vectors and the Economy of Indices

1.1.1 Upper and lower components

Greek indices run over all four spacetime coordinates, while Latin indices run only over the three spatial coordinates:

$$\mu, \nu, \rho, \sigma = 0, 1, 2, 3, \quad m, n = 1, 2, 3. \quad (1.1)$$

A generic contravariant four-vector is $A^\mu = (A^0, A^1, A^2, A^3)$. With the convention used here,

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad A_\mu = \eta_{\mu\nu} A^\nu. \quad (1.2)$$

Thus lowering an index changes the sign of the time component and leaves the spatial components alone:

$$A_0 = -A^0, \quad A_m = A^m. \quad (1.3)$$

For flat spacetime this is all the operation amounts to, although the same notation will acquire a richer geometric meaning in general relativity.

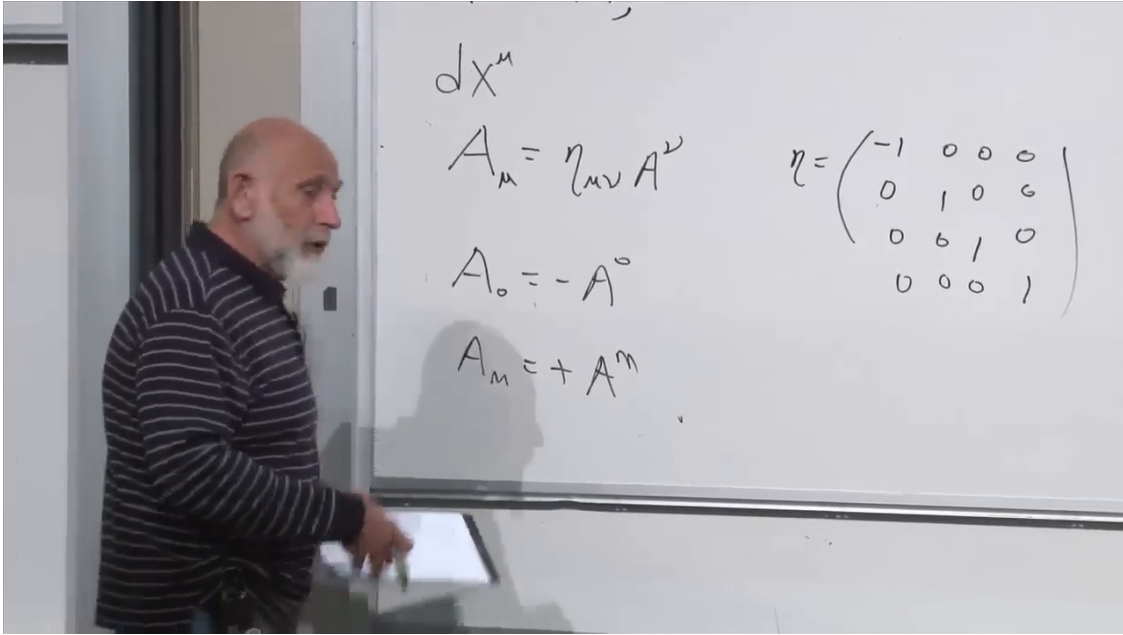


Figure 1.1: The Minkowski metric and the component rule for lowering a four-vector index.

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Einstein's summation convention is used when an index is repeated once upstairs and once downstairs. The contraction

$$A_\mu B^\mu = -A^0 B^0 + A^1 B^1 + A^2 B^2 + A^3 B^3 \quad (1.4)$$

is therefore a scalar. If a sum does not have this upper-lower repeated form, we shall display its summation sign explicitly.

1.1.2 The derivative carries a lower index

The four derivative operators are collected into

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu}. \quad (1.5)$$

Acting on a scalar field $\phi(x)$, they form the covariant vector $\partial_\mu \phi$. The quickest way to see the index placement is to consider an infinitesimal displacement:

$$d\phi = \frac{\partial \phi}{\partial x^\mu} dx^\mu = \partial_\mu \phi dx^\mu. \quad (1.6)$$

The left-hand side is the difference of scalar values at neighboring events, hence a scalar. Since dx^μ is contravariant, the object contracted with it must be covariant.

Question from the audience. If x^μ has an upper index, why does differentiation with respect to x^μ produce a lower index?

Response. Because the differential change $d\phi = \partial_\mu \phi dx^\mu$ is a scalar. A contravariant displacement can produce that scalar only by contracting with a covariant gradient. The position of the derivative index is therefore fixed by its transformation law, not by the typography of the fraction alone. ^a

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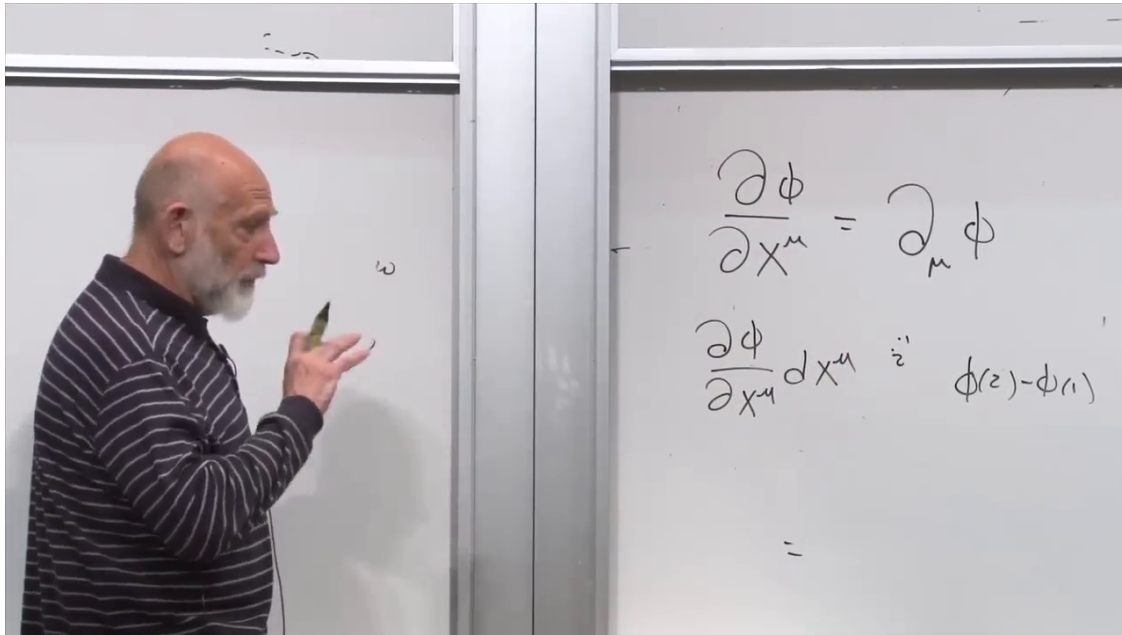


Figure 1.2: The scalar differential $d\phi$ identifies $\partial_\mu \phi$ as a covariant vector.

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The same rule provides another scalar. If $B^\mu(x)$ is a four-vector field, then

$$\partial_\mu B^\mu \quad (1.7)$$

is its four-divergence. It combines the time derivative of the time component with the three spatial derivatives of the corresponding spatial components.

All of this has a three-dimensional analogue. In Euclidean three-space the metric is the identity matrix, so raising or lowering a Latin index changes nothing. We shall consequently move spatial indices up and down when it is convenient, while continuing to distinguish spacetime indices carefully.

1.2 Lorentz Transformations in Matrix Form

Scalars and vectors are defined by the way they transform. A scalar keeps the same value, whereas a contravariant four-vector obeys

$$(A^\mu)' = L^\mu_\nu A^\nu. \quad (1.8)$$

The visible index μ is free and occurs on both sides. The repeated index ν is summed and disappears from the final component equation.

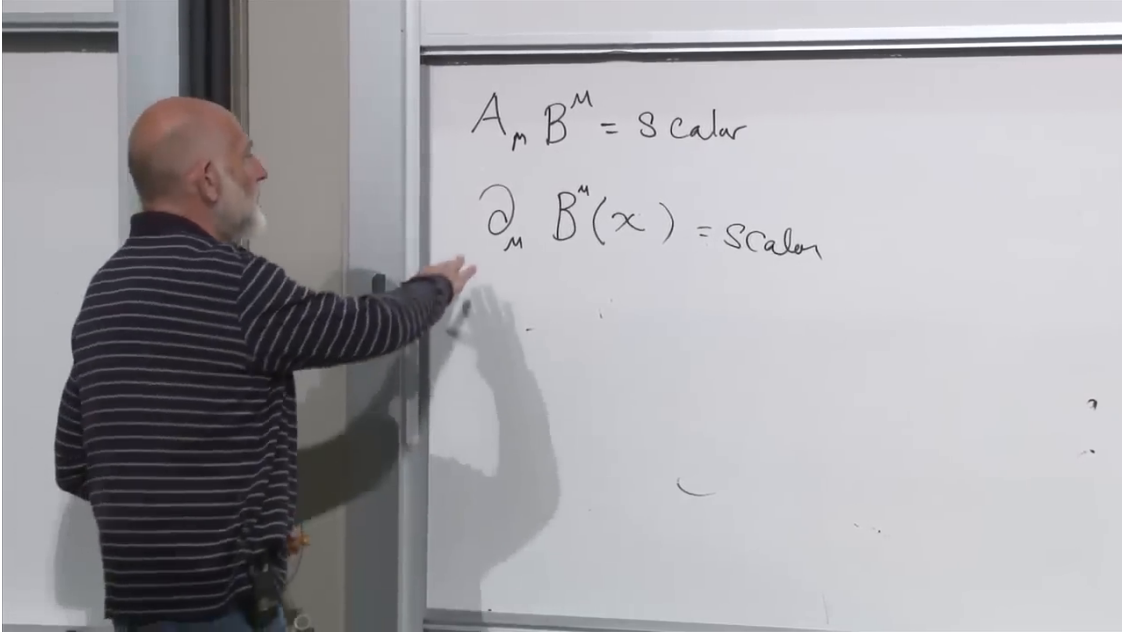
Here *Lorentz transformation* includes spatial rotations as well as boosts. A boost may point along any axis; rotations can be composed with boosts to produce a general transformation. A spatial rotation itself does not alter the time coordinate.

Question from the audience. When we rotate space, is time left completely unaffected?

Response. Yes. In an ordinary spatial rotation the time component is unchanged. Such rotations are included in the Lorentz group, alongside transformations that mix time with space.

^a

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Figure 1.3: Two scalar contractions: $A_\mu B^\mu$ and the four-divergence $\partial_\mu B^\mu(x)$.

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For a boost of speed v along the x -axis, with $c = 1$,

$$L^\mu{}_\nu = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \gamma = \frac{1}{\sqrt{1-v^2}}. \quad (1.9)$$

Multiplication by $(t, x, y, z)^\top$ gives

$$t' = \gamma(t - vx), \quad x' = \gamma(x - vt), \quad y' = y, \quad z' = z. \quad (1.10)$$

The class supplies the zeros and the two unit entries as the matrix is completed on the board.

A rotation through angle θ in the y - z plane instead has

$$R_{yz}(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \theta & \sin \theta \\ 0 & 0 & -\sin \theta & \cos \theta \end{pmatrix}. \quad (1.11)$$

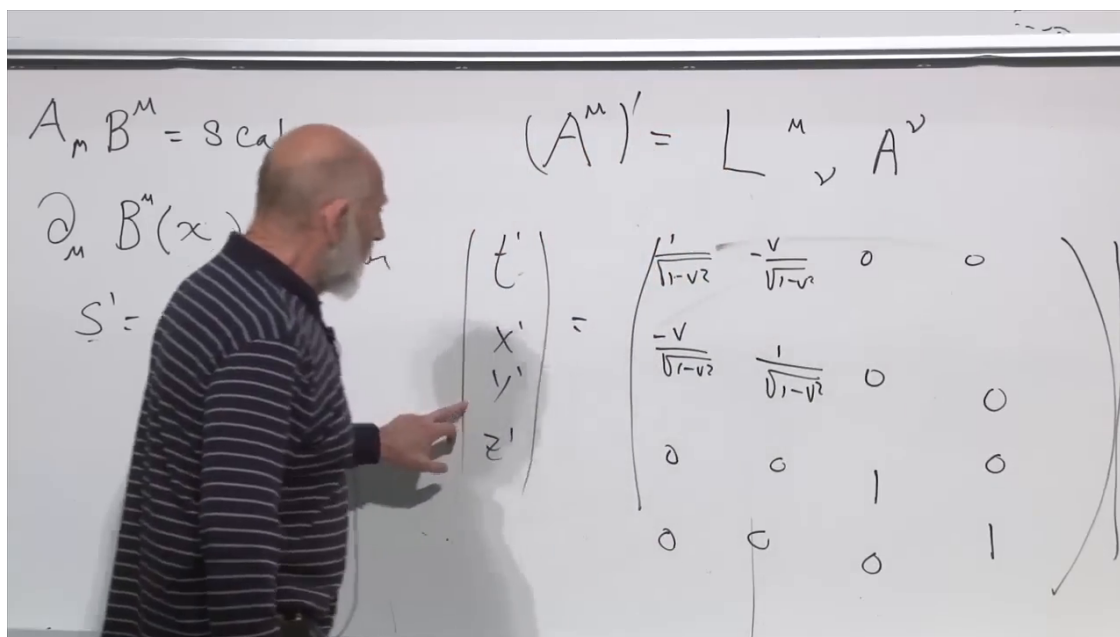
It says

$$y' = y \cos \theta + z \sin \theta, \quad z' = -y \sin \theta + z \cos \theta, \quad (1.12)$$

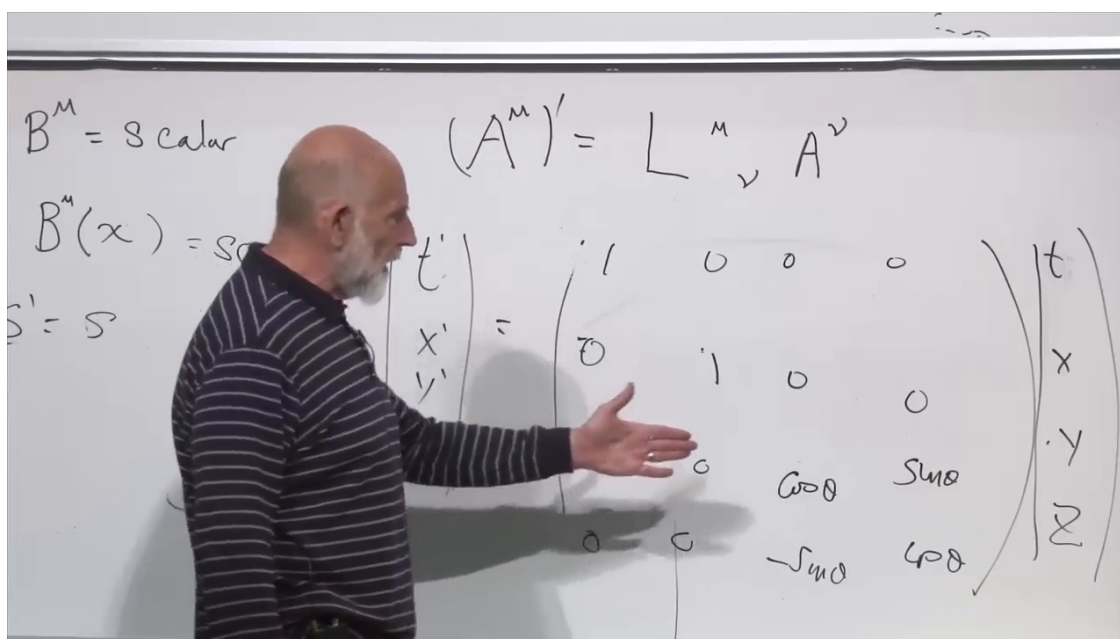
while t and x remain fixed. Products of matrices such as (1.9) and (1.11) generate more general Lorentz transformations.

A covariant vector transforms with a matrix carrying the opposite index placement:

$$(A_\mu)' = M_\mu{}^\nu A_\nu. \quad (1.13)$$

Figure 1.4: The x -directed boost matrix acting on the column $(t, x, y, z)^T$.

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Figure 1.5: A pure rotation in the y - z plane, with t and x untouched.

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An audience correction at the board catches a temporarily repeated free index: the unsummed μ on the left must also be the unsummed index on the right. The two matrices represent the same physical change of frame and are related by

$$M = \eta L \eta, \quad \eta^{-1} = \eta. \quad (1.14)$$

The last identity follows immediately because every diagonal entry of η is its own inverse.

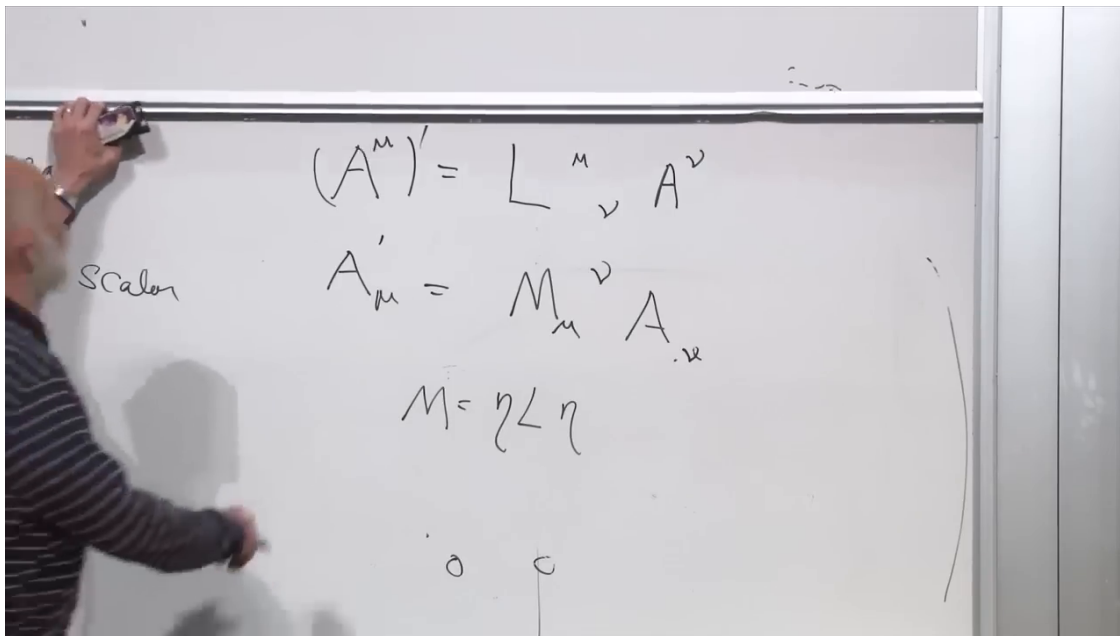


Figure 1.6: Contravariant and covariant transformation laws, related by $M = \eta L \eta$.

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The central idea is now available: a four-vector is an object whose components transform in parallel with the spacetime coordinates.

1.3 Tensors: One Transformation for Each Index

1.3.1 From vector products to the definition

A scalar is a rank-zero tensor, a vector is a rank-one tensor, and an object with two indices is a rank-two tensor. Begin with two vectors and form

$$T^{\mu\nu} = A^\mu B^\nu. \quad (1.15)$$

There are $4 \times 4 = 16$ components: $A^0 B^0, A^0 B^1, \dots, A^3 B^3$. The symbol $T^{\mu\nu}$ denotes this entire array, not one number.

Transforming each vector separately gives

$$(A^\mu)' = L^\mu_\sigma A^\sigma, \quad (1.16)$$

$$(B^\nu)' = L^\nu_\tau B^\tau, \quad (1.17)$$

and therefore

$$(T^{\mu\nu})' = L^\mu{}_\sigma L^\nu{}_\tau T^{\sigma\tau}. \quad (1.18)$$

Each tensor index receives its own Lorentz matrix. The dummy indices σ and τ may be renamed at will, provided each pair remains consistent.

Question from the audience. Does this rearrangement assume that the matrices and vector components commute?

Response. These are ordinary classical numbers, not noncommuting quantum operators, so they commute. Moreover, the same Lorentz matrix is acting on both indices. There is no ordering issue in this calculation. ^a

^aLecture timestamp: 00:31:56.

The rule extends immediately to rank three:

$$(T^{\mu\nu\lambda})' = L^\mu{}_\sigma L^\nu{}_\tau L^\lambda{}_\kappa T^{\sigma\tau\kappa}, \quad (1.19)$$

and similarly to any rank. This transformation law, rather than a particular factorization into vectors, is the definition of a tensor. For example,

$$A^\mu B^\nu + C^\mu D^\nu \quad (1.20)$$

is still a tensor even though it cannot generally be written as one product of two vectors.

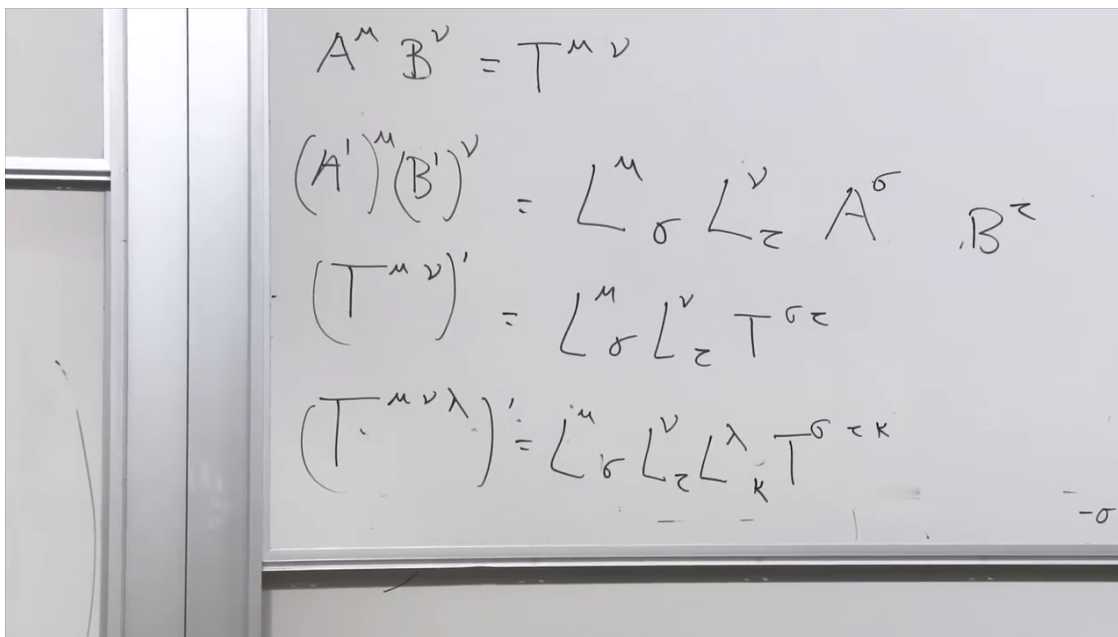


Figure 1.7: The board ladder from a product of vectors to rank-two and rank-three tensor transformation laws.

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Question from the audience. What does matrix multiplication by a rank-two tensor mean component by component?

Response. Fix the two output indices and sum over the two hidden indices. For example,

$$(T^{31})' = L^3_{\sigma} L^1_{\tau} T^{\sigma\tau}.$$

Since both σ and τ take four values, this one output component contains sixteen terms. ^a

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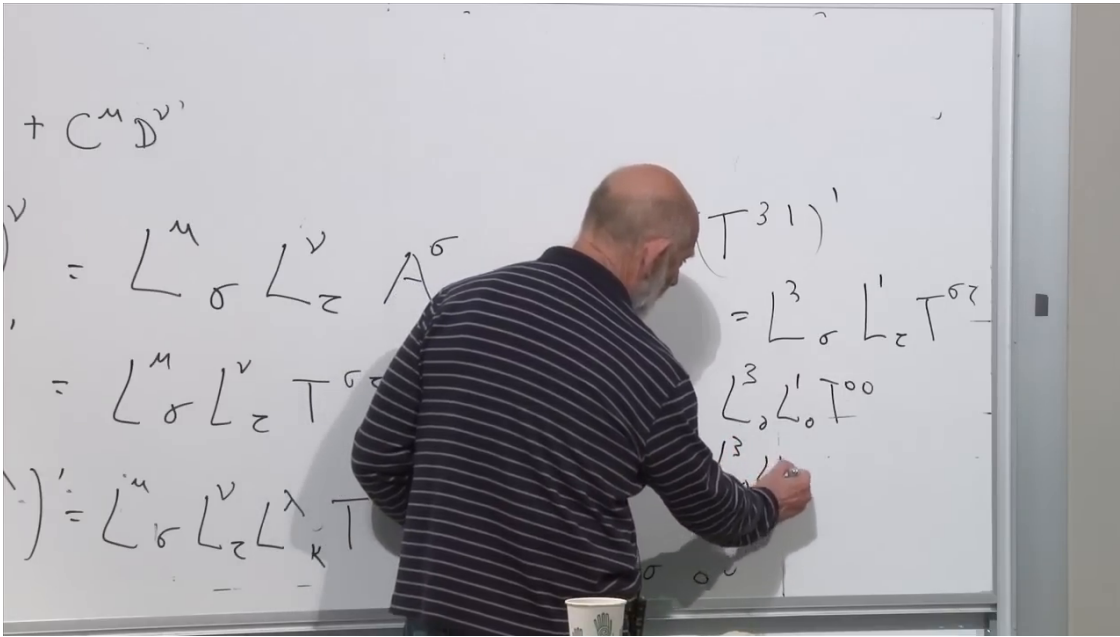


Figure 1.8: Expanding the fixed component $(T^{31})'$ makes the two independent summations explicit.

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Tensor equations have a decisive advantage. If two tensors have equal components in one frame, the transformation law makes them equal in every frame. Equivalently, if every component of a tensor vanishes in one frame, the whole tensor vanishes in every frame. One vanishing component is not enough; all components must vanish. This is why an equation written between tensors transports its truth from one inertial frame to another.

1.3.2 Raising and lowering tensor indices

An index of a tensor is lowered exactly as an index of a vector:

$$T^{\mu}_{\nu} = T^{\mu\sigma} \eta_{\sigma\nu}. \quad (1.21)$$

For the diagonal metric in (1.2), every lowered time index contributes one minus sign. Consequently,

$$T_{00} = T^{00}, \quad T_{01} = -T^{01}. \quad (1.22)$$

The first equality contains two sign changes; the second contains only one.

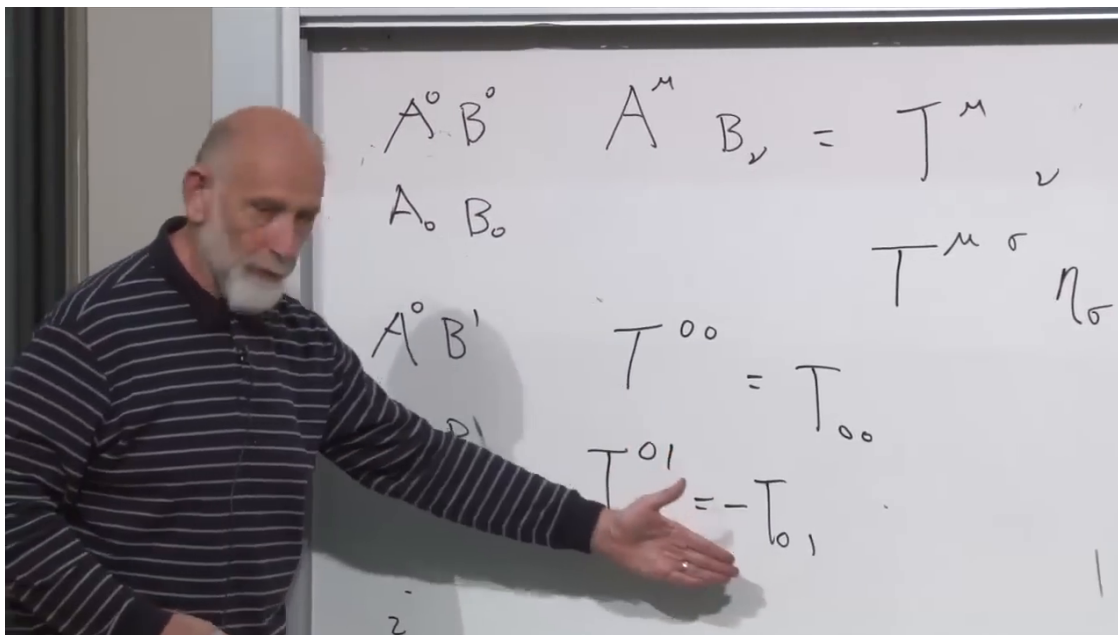


Figure 1.9: Examples of lowering two tensor indices: two time indices give two minus signs, while one time index gives one.

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Question from the audience. Why is it useful to keep both covariant and contravariant forms? Is there more to it than compact notation?

Response. For the present calculation, the immediate benefit is compact contraction and automatic management of the Minkowski signs. There is a deeper geometric distinction between vectors and covectors, but it is better developed in general relativity. In flat spacetime the index rules are so efficient that one usually works with them directly. ^a

^aLecture timestamp: 00:41:26.

Question from the audience. Does a tensor always have four components, or is that peculiar to relativity?

Response. The range of each index is set by the dimension of the space. A rank-one tensor in four-dimensional spacetime has four components; a rank-two tensor has sixteen. In three dimensions the corresponding counts are three and nine. The tensor rule itself is dimension-independent. ^a

^aLecture timestamp: 00:42:55.

Question from the audience. Can covariant and contravariant objects be described more geometrically?

Response. Yes, they have a geometric interpretation, but pursuing it now would postpone electrodynamics even further. For this course the transformation and index-manipulation rules are sufficient; the geometric distinction will become unavoidable in general relativity. ^a

^aLecture timestamp: 00:43:51.

1.4 Symmetry, Antisymmetry, and the Electromagnetic Tensor

For a generic rank-two tensor, exchanging the indices changes the component:

$$T^{\mu\nu} \neq T^{\nu\mu}. \quad (1.23)$$

Indeed, $A^\mu B^\nu$ and $A^\nu B^\mu$ are different in general; the specific components $A^0 B^1$ and $A^1 B^0$ need not agree.

There are two important special cases. A symmetric tensor satisfies

$$T^{\mu\nu} = T^{\nu\mu}, \quad (1.24)$$

and a concrete symmetric combination is

$$A^\mu B^\nu + A^\nu B^\mu. \quad (1.25)$$

An antisymmetric tensor satisfies

$$F^{\mu\nu} = -F^{\nu\mu}, \quad (1.26)$$

as does the combination

$$A^\mu B^\nu - A^\nu B^\mu. \quad (1.27)$$

Setting the indices equal in (1.26) gives

$$F^{00} = -F^{00}, \quad F^{00} = 0, \quad (1.28)$$

and the same argument removes every diagonal component. A 4×4 antisymmetric tensor therefore has six independent components.

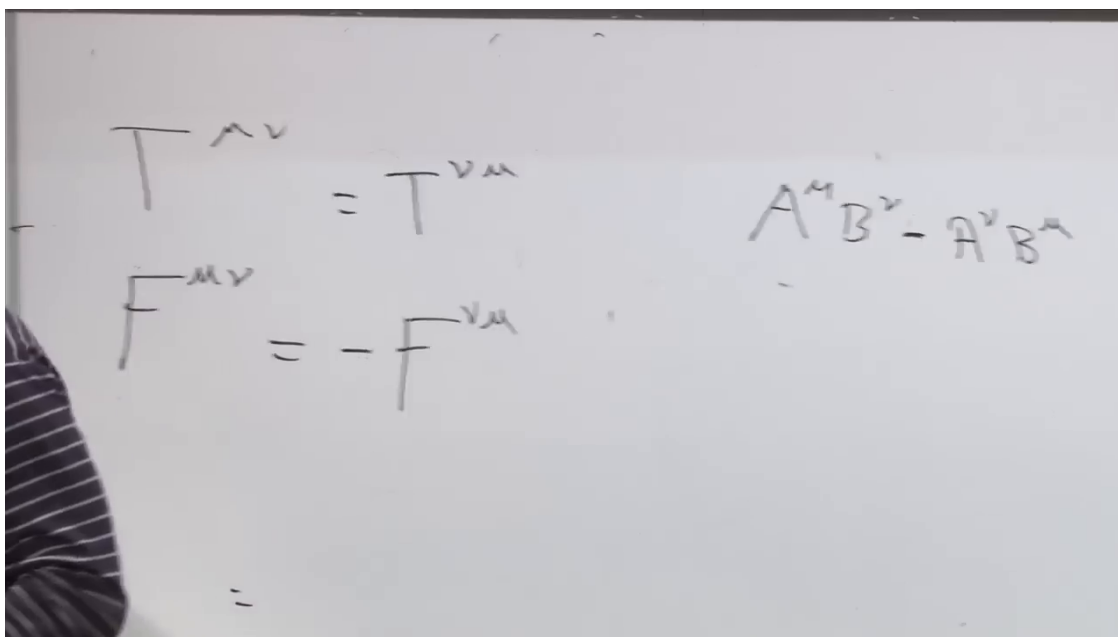


Figure 1.10: Symmetric and antisymmetric rank-two tensors compared on the board.

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Those six components are precisely enough for the three components of an electric field and the three components of a magnetic field. With the sign convention written on the board,

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & -B_3 & B_2 \\ E_2 & B_3 & 0 & -B_1 \\ E_3 & -B_2 & B_1 & 0 \end{pmatrix}. \quad (1.29)$$

Reflection across the diagonal reverses every sign. Electric and magnetic fields are therefore not independent relativistic objects. A boost can mix them just as it mixes space and time: a field that is purely electric in one frame may have a magnetic component in another.

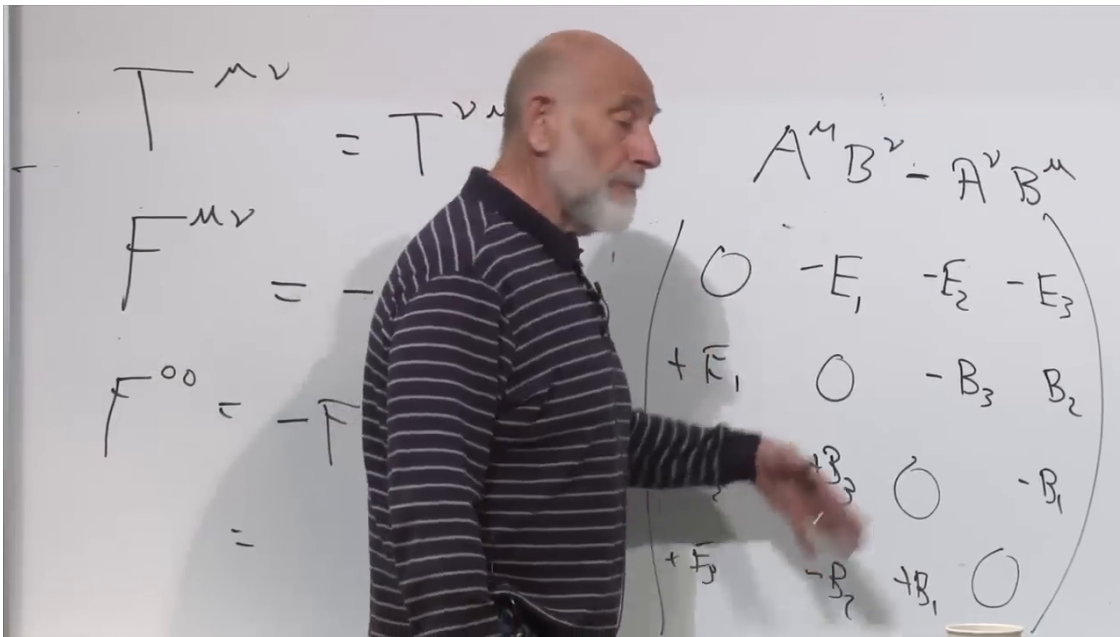


Figure 1.11: The completed antisymmetric array whose six independent entries are **E** and **B**.

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The board carrying (1.29) remains visible while the discussion moves to particle dynamics. We now reuse the symbol A_μ , not for the generic vector of the preceding examples, but for the electromagnetic four-potential.

1.5 The Charged-Particle Action

There are two possible starting points. We could study the dynamics of the field itself, which will lead to Maxwell's equations, or study a charged particle moving in a prescribed field, which leads to the Lorentz force. We begin with the particle.

At low velocity the target is

$$m\mathbf{a} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (1.30)$$

with a factor $1/c$ multiplying the magnetic term in the Gaussian-unit version. We set $c = 1$. The two terms in (1.30) look different in three-vector notation, but relativity should reveal them as parts of one object.

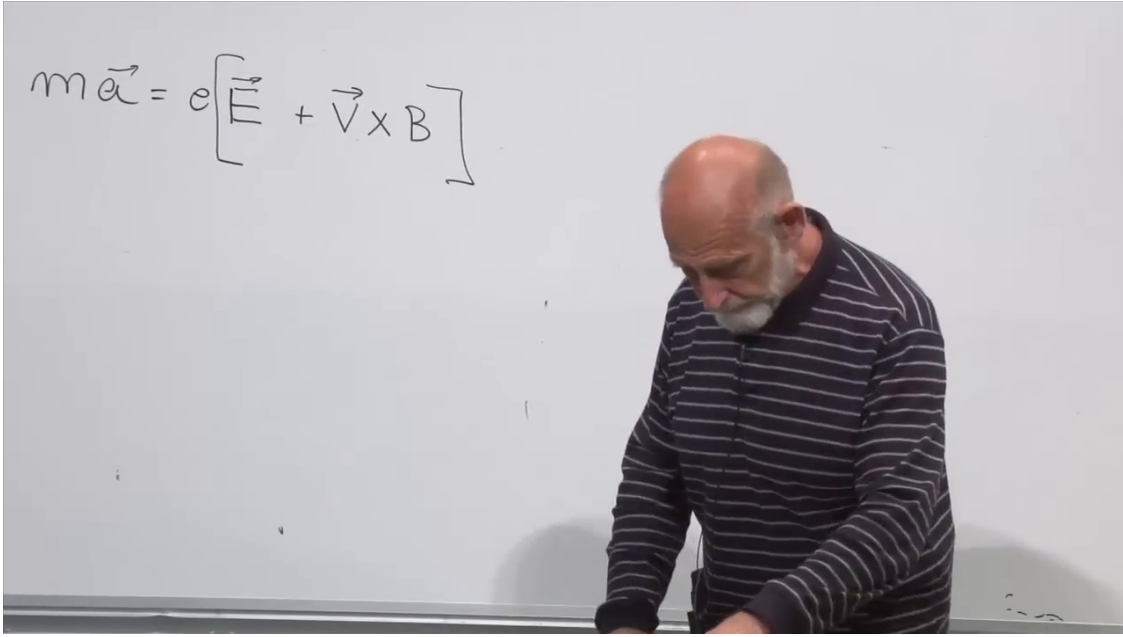


Figure 1.12: The nonrelativistic Lorentz force, the result the invariant action must reproduce.

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1.5.1 Free motion and the vector potential

For a free relativistic particle,

$$S_{\text{free}} = -m \int_1^2 d\tau = \int_{t_1}^{t_2} dt \left[-m \sqrt{1 - \dot{\mathbf{x}}^2} \right], \quad (1.31)$$

where

$$\dot{\mathbf{x}}^2 = (\dot{x}^1)^2 + (\dot{x}^2)^2 + (\dot{x}^3)^2. \quad (1.32)$$

The quantity in square brackets is the free-particle Lagrangian. Its momentum and energy were found earlier; we retain the action and ask what field-dependent term may be added.

The basic electromagnetic variable is the four-vector potential $A_\mu(x)$. The electric and magnetic fields will be derived from it. Along a small worldline segment dx^μ , the contraction

$$A_\mu(x) dx^\mu \quad (1.33)$$

is a scalar. Adding those scalar increments from one endpoint to the other gives a Lorentz-invariant interaction:

$$S_{\text{int}} = -e \int_1^2 A_\mu(x) dx^\mu. \quad (1.34)$$

The coefficient e is the electric charge. The overall sign belongs to the historical charge convention; changing the convention changes the sign assigned to e , not the content of the theory.

1.5.2 Ordinary-time form

To use the familiar Euler–Lagrange equations, divide and multiply (1.34) by dt . Since

$$\frac{dx^0}{dt} = 1, \quad \frac{dx^m}{dt} = \dot{x}^m, \quad (1.35)$$

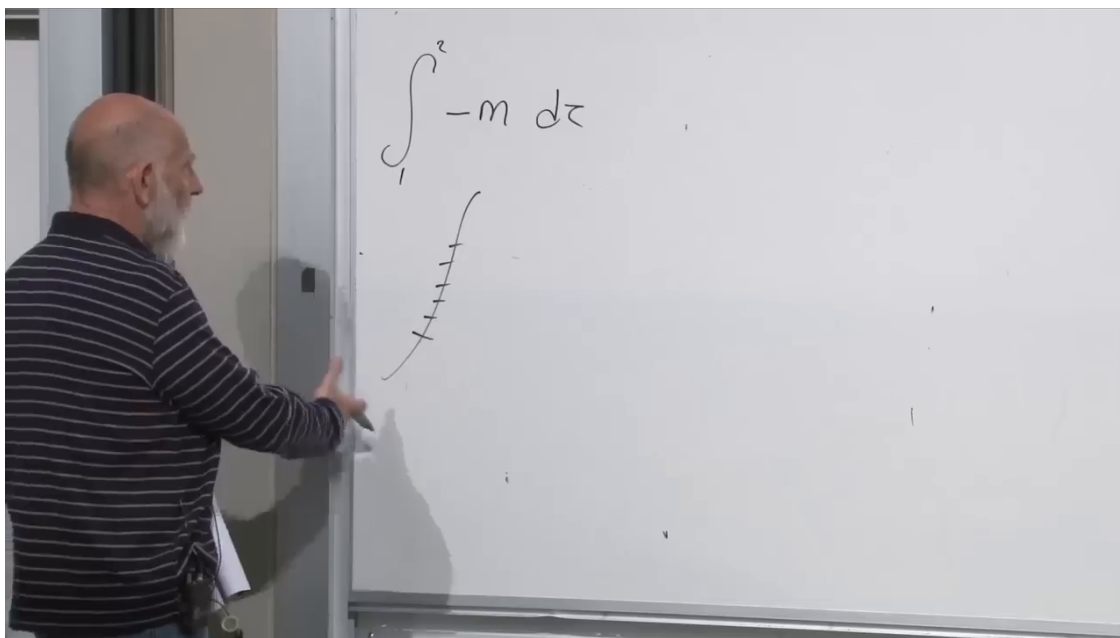


Figure 1.13: The free action as proper time accumulated along a worldline.

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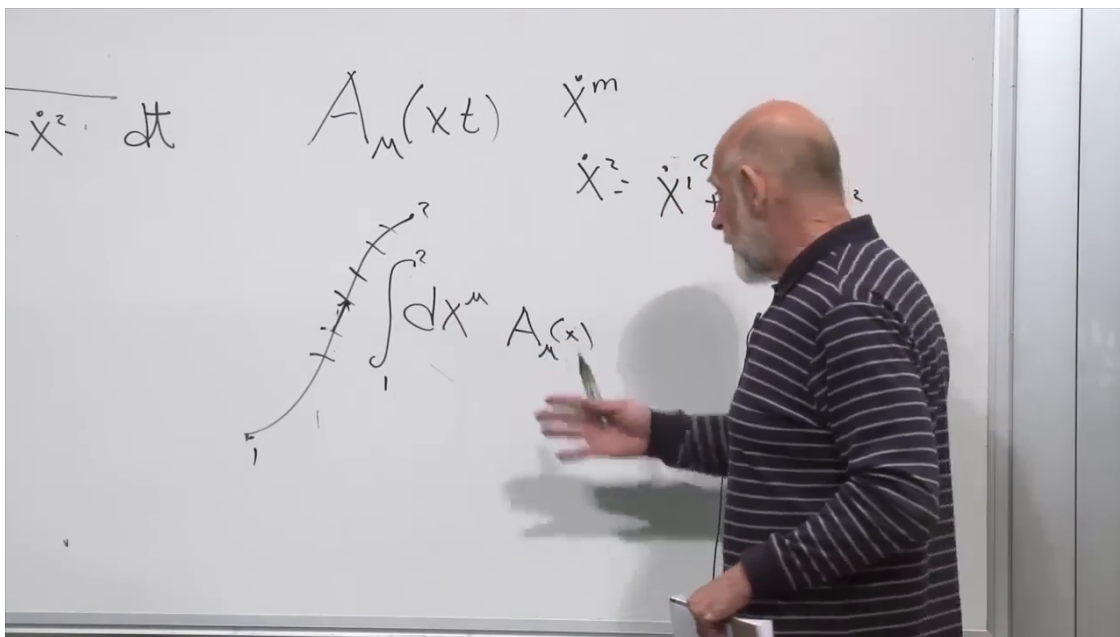


Figure 1.14: A worldline segment contracted with the four-potential to form the scalar increment $A_\mu dx^\mu$.

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the complete action is

$$S = \int dt \left[-m\sqrt{1 - \dot{\mathbf{x}}^2} - eA_0(x, t) - e\dot{x}^m A_m(x, t) \right]. \quad (1.36)$$

Thus

$$\boxed{L = -m\sqrt{1 - \dot{\mathbf{x}}^2} - eA_0 - e\dot{x}^m A_m}. \quad (1.37)$$

For now $A_\mu(x, t)$ is a known external field. We vary the worldline, not the field.

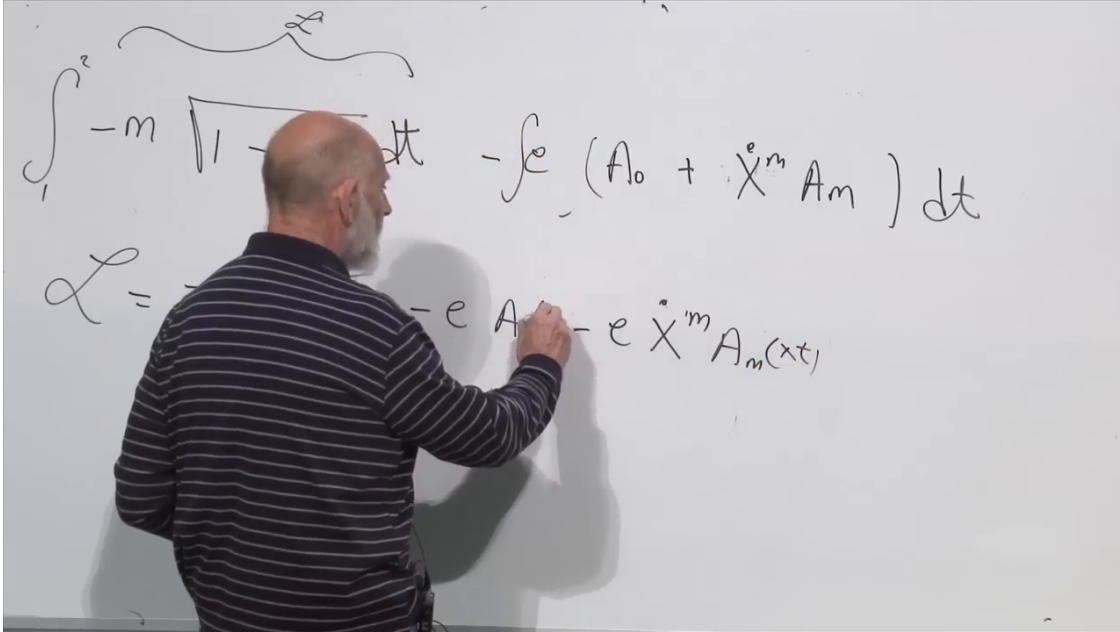


Figure 1.15: The invariant worldline coupling rewritten as an ordinary-time Lagrangian.

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1.6 Grinding Out the Lorentz Force

For a spatial component x^m , the Euler–Lagrange equation reads

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}^m} = \frac{\partial L}{\partial x^m}. \quad (1.38)$$

Differentiating with respect to velocity gives

$$\frac{\partial L}{\partial \dot{x}^m} = \frac{m\dot{x}_m}{\sqrt{1 - \dot{\mathbf{x}}^2}} - eA_m. \quad (1.39)$$

The letter m unfortunately serves both as the mass and as a spatial index. Context distinguishes them: the first m in (1.39) is mass; the subscript m labels one of the three spatial components.

Question from the audience. Is the first attempt at the left-hand side missing a term?

Response. Yes. Differentiating only the free-particle term omits the velocity dependence of $-e\dot{x}^m A_m$. The full derivative is $m\dot{x}_m/\sqrt{1 - \dot{\mathbf{x}}^2} - eA_m$. The correction is essential because its time derivative supplies part of both the electric and magnetic force. ^a

^aLecture timestamp: 01:08:18.

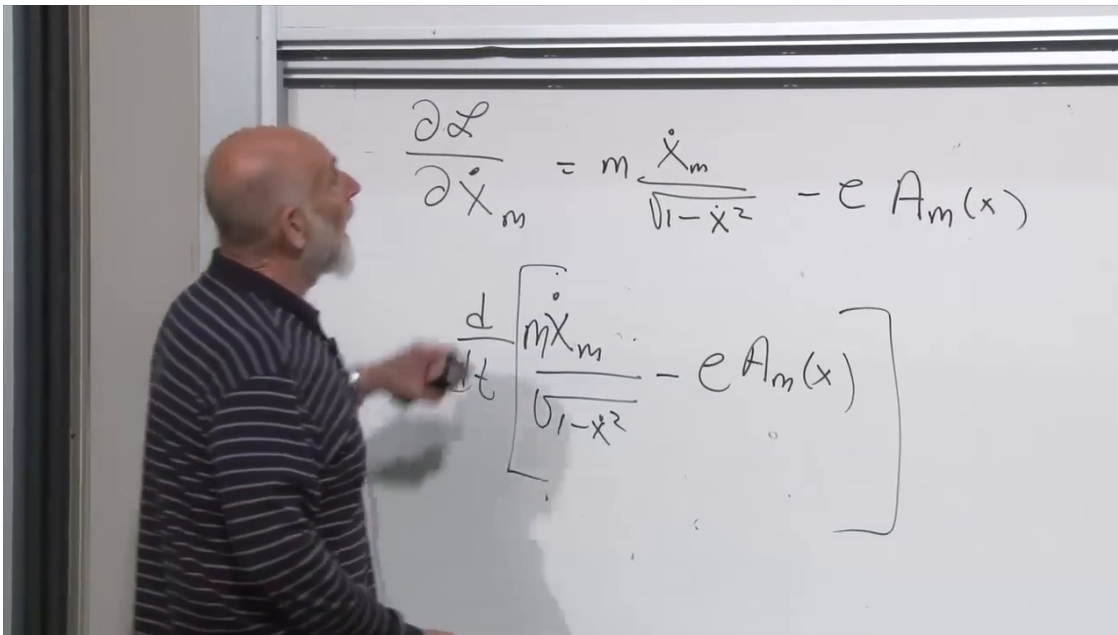


Figure 1.16: The momentum derivative contains both the mechanical term and the vector-potential term.

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The right-hand side of (1.38) is

$$\frac{\partial L}{\partial x^m} = -e \partial_m A_0 - e \dot{x}^n \partial_m A_n. \quad (1.40)$$

Therefore

$$\frac{d}{dt} \left(\frac{m \dot{x}_m}{\sqrt{1-\dot{\mathbf{x}}^2}} - e A_m \right) = -e \partial_m A_0 - e \dot{x}^n \partial_m A_n. \quad (1.41)$$

Question from the audience. Have covariant and contravariant spatial components been interchanged in this calculation?

Response. They have been moved up and down, but for Latin spatial indices the Euclidean spatial block of the metric is the identity. Thus $x_m = x^m$ and $A_m = A^m$. Only a time index produces the sign change. ^a

^aLecture timestamp: 01:09:29.

Question from the audience. Are repeated Latin indices summed too?

Response. Yes. A repeated Latin index is summed from 1 to 3; a repeated Greek index is summed from 0 to 3. A lone index such as the m labeling one spatial equation is free and is not summed. The mass symbol is not an index. ^a

^aLecture timestamp: 01:10:00.

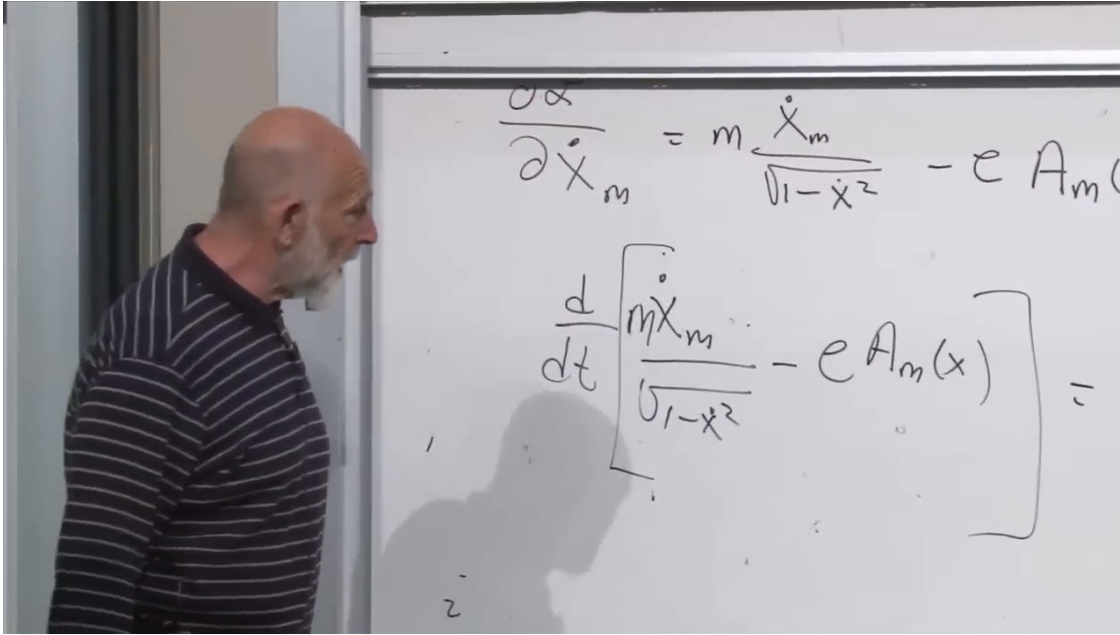


Figure 1.17: The Euler–Lagrange equation before the vector potential’s total time derivative is expanded.

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1.6.1 The chain rule along the moving worldline

Even a time-independent field changes along the particle’s path because the particle samples different positions. The total derivative is

$$\frac{dA_m(x(t), t)}{dt} = \partial_0 A_m + \dot{x}^n \partial_n A_m. \quad (1.42)$$

The first term is explicit time dependence; the second is implicit time dependence through $x^n(t)$.

Substituting (1.42) into (1.41) and moving the field terms to the right gives

$$m \frac{d}{dt} \left(\frac{\dot{x}_m}{\sqrt{1-\dot{\mathbf{x}}^2}} \right) = e (\partial_0 A_m - \partial_m A_0) + e \dot{x}^n (\partial_n A_m - \partial_m A_n). \quad (1.43)$$

The calculation has finally found its organizing principle. The first bracket contains no velocity; the second is linear in velocity. This is exactly the split between electric and magnetic forces.

Define the electric components in the convention of the derivation by

$$E_m = \partial_0 A_m - \partial_m A_0. \quad (1.44)$$

The spatial antisymmetric combination

$$C_{mn} \equiv \partial_n A_m - \partial_m A_n \quad (1.45)$$

contains the three independent components of the curl of the vector potential. Its orientation is chosen so that

$$\dot{x}^n C_{mn} = (\mathbf{v} \times \mathbf{B})_m. \quad (1.46)$$

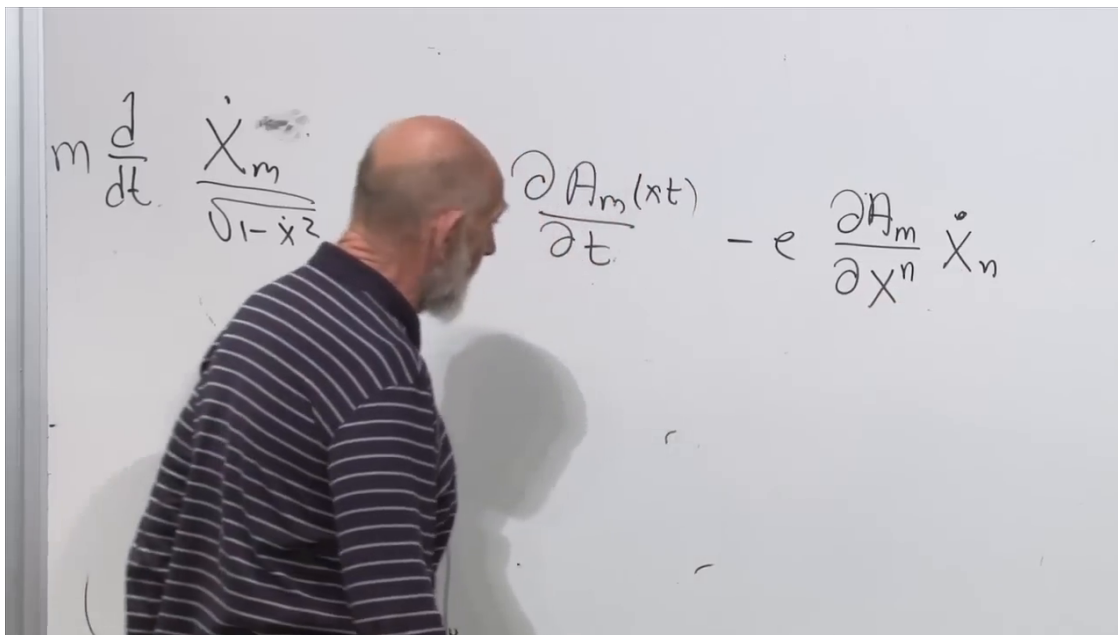


Figure 1.18: The vector potential changes explicitly with time and implicitly as the particle moves through space.

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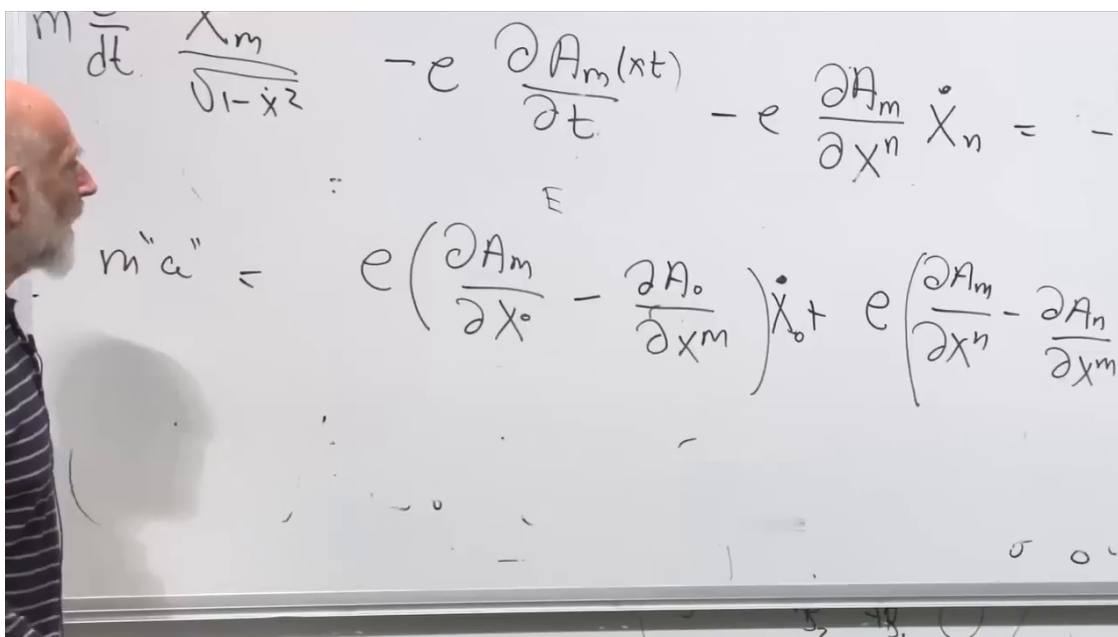


Figure 1.19: The force terms grouped into a velocity-independent electric part and a velocity-dependent antisymmetric part.

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Equation (1.43) is therefore the relativistic momentum form of (1.30). The factor under the time derivative is the relativistic momentum per unit mass, not the Newtonian velocity alone.

The derivation is worth repeating by hand: begin with (1.37), compute both sides of (1.38), apply the chain rule, and group terms according to their velocity dependence. The exercise is simultaneously practice in variational mechanics, multivariable calculus, and electrodynamics.

Question from the audience. Why choose $-e \int A_\mu dx^\mu$ as the electromagnetic interaction?

Response. Experiment ultimately decides the coupling, but two structural requirements make this the natural first candidate. The action is assembled from local increments along the trajectory, and each increment must be a scalar. The scalar $dx^\mu dx_\mu = d\tau^2$ is already represented by the free action. Contracting dx^μ with the field A_μ is the simplest new field-dependent scalar. More elaborate scalars can be written, and gauge invariance will later sharpen the choice. ^a

^aLecture timestamp: 01:25:34.

Question from the audience. Can Maxwell's equations also be derived from least action?

Response. Yes. The present action governs a particle in a given electromagnetic field. To obtain Maxwell's equations we must add the field's own action and vary the field itself, just as we did for the scalar field. ^a

^aLecture timestamp: 01:28:58.

Question from the audience. Where is the summation in the grouped force equation?

Response. The free m labels one of the three component equations and is not summed. The repeated n in $\dot{x}^n(\partial_n A_m - \partial_m A_n)$ is summed over 1, 2, 3. Written without Einstein notation, that term would carry an explicit $\sum_{n=1}^3$. ^a

^aLecture timestamp: 01:29:26.

Question from the audience. Why identify $\partial_0 A_m - \partial_m A_0$ as the electric field rather than assume the usual formula in advance?

Response. The action has produced a force term that depends on position and time but not on the particle velocity. That is precisely the operational role of the electric field. The remaining antisymmetric terms multiply the velocity and therefore have the magnetic role. One may regard this calculation as discovering the fields from the force law rather than inserting separate definitions beforehand. ^a

^aLecture timestamp: 01:31:05.

1.7 The Manifestly Covariant Force Law

The left-hand side of (1.43) is not yet written in an obviously four-dimensional form. Introduce the four-velocity

$$u^\mu = \frac{dx^\mu}{d\tau}. \quad (1.47)$$

Its spatial components obey

$$u^m = \frac{dx^m}{d\tau} = \frac{\dot{x}^m}{\sqrt{1 - \dot{\mathbf{x}}^2}}. \quad (1.48)$$

Multiplying the spatial equation by $dt/d\tau$ converts d/dt into $d/d\tau$:

$$\frac{dt}{d\tau} \frac{d}{dt} = \frac{d}{d\tau}. \quad (1.49)$$

Hence the left-hand side becomes the spatial part of the proper acceleration,

$$m \frac{d^2 x^m}{d\tau^2}. \quad (1.50)$$

The right-hand side can be written without separating time and space:

$$m \frac{d^2 x^m}{d\tau^2} = e (\partial_\nu A_m - \partial_m A_\nu) \frac{dx^\nu}{d\tau}. \quad (1.51)$$

For $\nu = 0$ this reproduces the electric bracket; for $\nu = 1, 2, 3$ it reproduces the magnetic bracket.

Both sides now have the transformation character of a four-vector. The three spatial equations are therefore components of

$$\boxed{m \frac{d^2 x^\mu}{d\tau^2} = e (\partial_\nu A^\mu - \partial^\mu A_\nu) u^\nu}. \quad (1.52)$$

If one defines

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (1.53)$$

then the index order and interaction sign used in this lecture give the equivalent compact form

$$m \frac{du^\mu}{d\tau} = -e F^\mu{}_\nu u^\nu. \quad (1.54)$$

1

Question from the audience. Why is it legitimate to add a fourth, time-component equation? Is $d^2 x^0/d\tau^2$ simply zero?

Response. It is not generally zero: $x^0 = t$, but $dt/d\tau = \gamma$ changes when the particle's speed changes. The decisive argument is covariance. The scalar action guarantees Lorentz-invariant equations; once the spatial components are recognized as components of one four-vector equation, Lorentz transformations mix them with the time component. The fourth equation must accompany the other three. ^a

^aLecture timestamp: 01:42:40.

1.7.1 What the fourth component means

The three spatial components are the relativistic statement

$$\frac{d\mathbf{p}}{dt} = \mathbf{F}. \quad (1.55)$$

¹Editorial clarification: Electrodynamics texts distribute signs differently among the metric signature, the definition of A_μ , the charge, and $F_{\mu\nu}$. Equations (1.52) and (1.54) retain the sign and index order obtained from the action $-e \int A_\mu dx^\mu$; the invariant content is the antisymmetrized derivative of the potential.

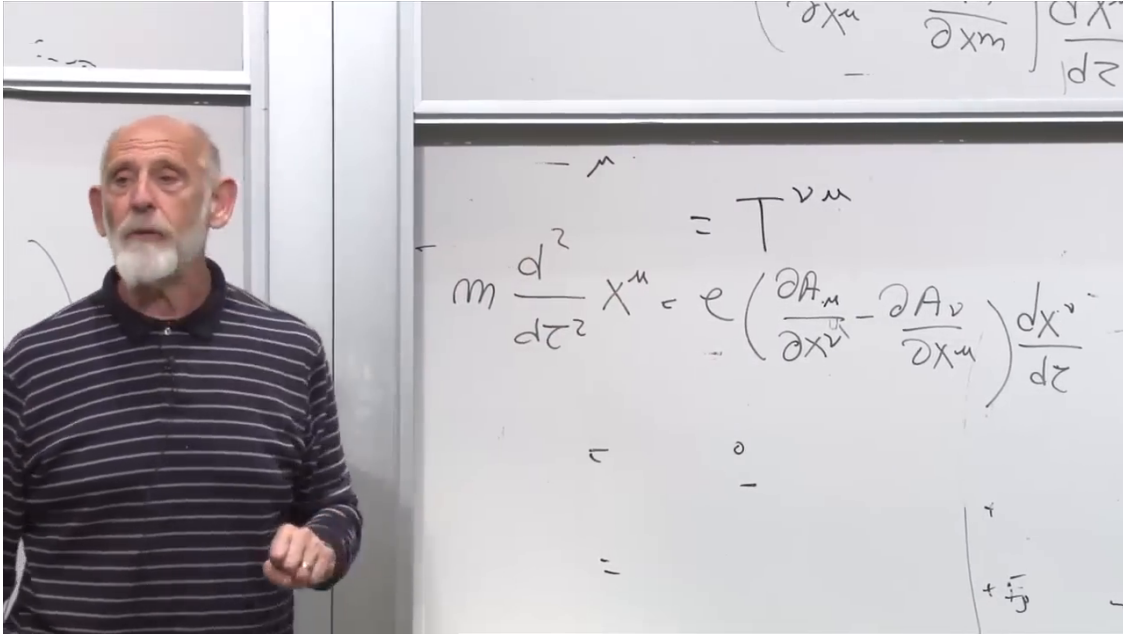


Figure 1.20: The three spatial equations assembled into a four-vector equation using proper time.

Lecture frame: 01:42:40.

Energy and momentum are themselves the components of a four-vector, so the time component must describe the corresponding energy balance. In three-vector language,

$$\frac{dK}{dt} = \mathbf{F} \cdot \mathbf{v}. \quad (1.56)$$

It records the rate at which the field does work on the particle. The magnetic contribution is perpendicular to $\mathbf{v}$ and does no work; the electric contribution changes the particle energy.

Question from the audience. If covariance supplies the fourth equation formally, how do we know it has physical meaning?

Response. The first three components say that force changes momentum. The fourth says that work changes energy: $dK/dt = \mathbf{F} \cdot \mathbf{v}$. Since energy and momentum form one four-vector, their response to the field must also be one four-component statement. ^a

^aLecture timestamp: 01:45:20.

The method is more important than the final compression. We required the action to be a scalar before varying it. The resulting equations could then be reorganized into tensors without repairing their symmetry after the fact. In modern field theory this becomes a general design rule: construct the Lagrangian to respect the symmetries of the problem.

1.8 Locality, Lorentz Invariance, and the Next Symmetry

Three principles recur throughout relativistic field theory. Two have already been used; the third will be developed next.

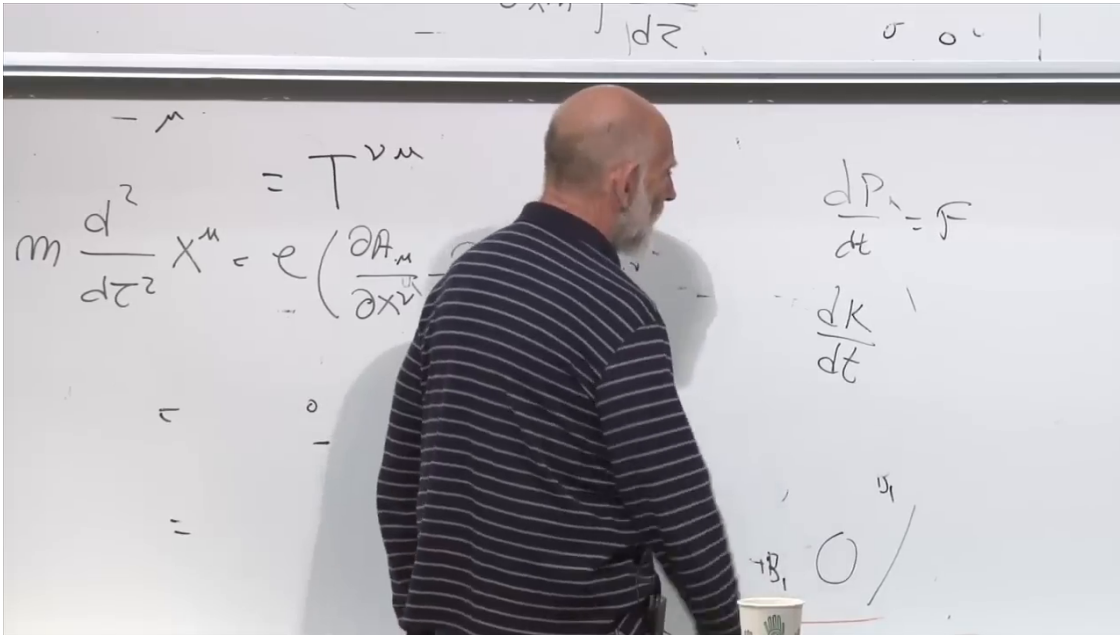


Figure 1.21: The spatial law $d\mathbf{p}/dt = \mathbf{F}$ and its time-component partner, the work-energy relation.

Lecture frame: 01:47:10.

1.8.1 Locality

Locality says that the change of a particle or field is controlled by what happens infinitesimally nearby in spacetime. A particle responds to the field evaluated on its worldline, not to an unrelated value far away. A field at one point responds to the field and its derivatives in a neighborhood of that point, not directly to arbitrary remote values or to future data.

For a field, the action therefore takes the local form

$$S = \int d^3x dt \mathcal{L}(\phi, \partial_\mu \phi). \quad (1.57)$$

The integral adds one contribution from each spacetime point. First derivatives encode comparison with neighboring points, and the variational principle produces differential equations. One could imagine nonlocal products coupling fields at widely separated points, but their equations would make the response here depend directly on conditions there. That is not the local structure being imposed. Actions with higher derivatives are possible and form an interesting subject, but they are not needed for the present construction.

Question from the audience. Does $\partial_\mu \phi$ represent four derivative components?

Response. Yes. It denotes the time derivative and the three spatial derivatives of ϕ , collected as one covariant four-vector. ^a

^aLecture timestamp: 01:52:25.

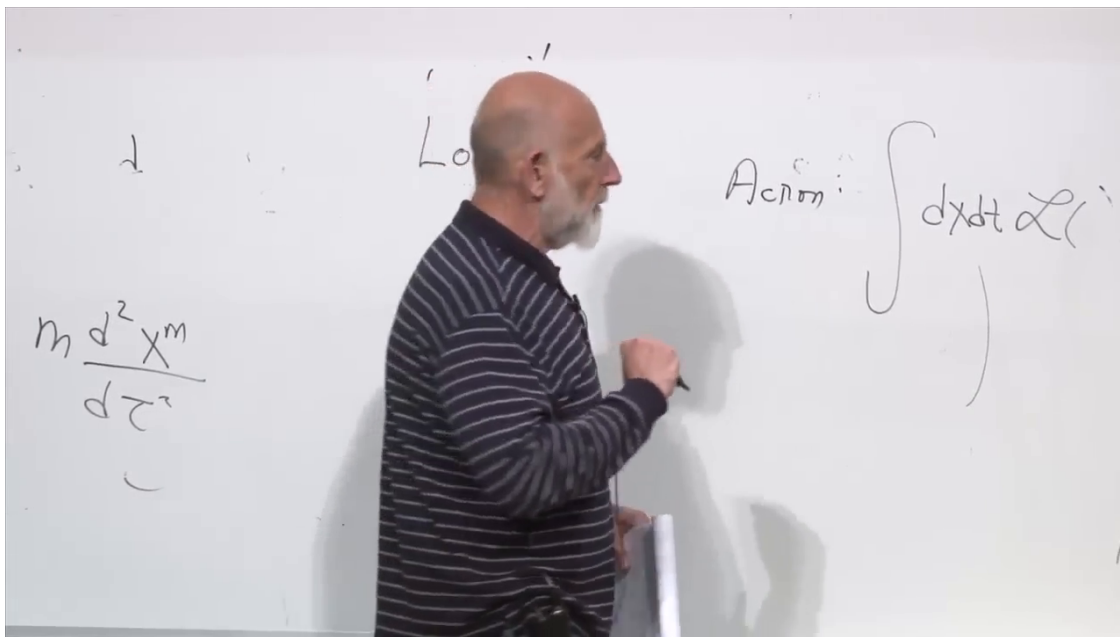


Figure 1.22: Locality expressed as a spacetime integral of a density built from fields and their nearby derivatives.

Lecture frame: 01:51:00.

Question from the audience. How is *nearby* quantified in the locality principle?

Response. By derivatives. A first derivative compares the field at a point with its value at infinitesimally neighboring points. A local Lagrangian depends on the fields and a finite set of such derivatives at the same spacetime point. ^a

^aLecture timestamp: 01:52:55.

Question from the audience. How does the field version of locality translate to the particle calculation just completed?

Response. The particle Lagrangian is local along the worldline: it depends on neighboring worldline positions through the velocity and on the electromagnetic potential evaluated at the particle's current position. The particle's acceleration here does not depend directly on the field at a distant point. ^a

^aLecture timestamp: 01:53:31.

1.8.2 Lorentz invariance

Lorentz invariance is implemented by building the Lagrangian from scalars. Observers may decompose those scalars into different time and space components, but they agree on the action. The tensor equations obtained by variation consequently have the same form in every inertial frame.

1.8.3 Gauge invariance

Gauge invariance is the remaining principle. It is announced but not yet defined here. In the next step it will constrain which features of the four-potential are physically meaningful and will guide the field action that produces Maxwell's equations.

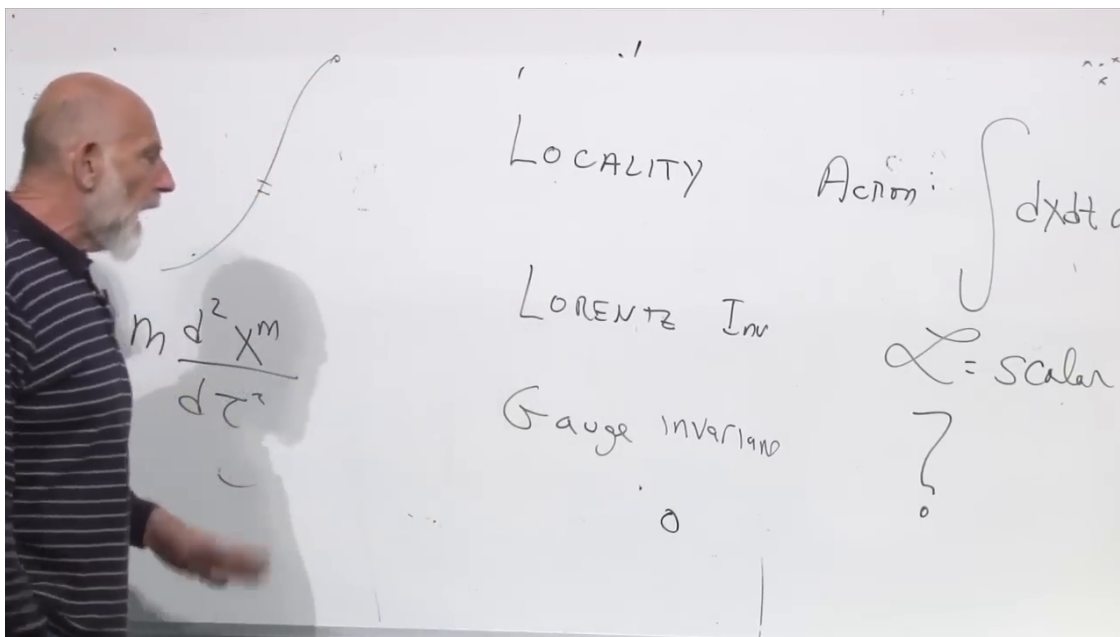


Figure 1.23: The three organizing principles: locality, Lorentz invariance, and gauge invariance.

Lecture frame: 01:55:40.

These principles recur in general relativity, quantum electrodynamics, Yang–Mills theory, and the Standard Model. Details such as the number of particle species vary from theory to theory; locality and symmetry provide the deeper framework. The practical assignment is therefore to retrace the Euler–Lagrange calculation until every index and every grouped term can be accounted for. Gauge invariance will then enter as the next constraint on the same electromagnetic action.